

THE AI / ML TIMELINE

From logic machines to generative intelligence

FOUNDATIONS

AI WINTERS

DEEP LEARNING

GENERATIVE AI



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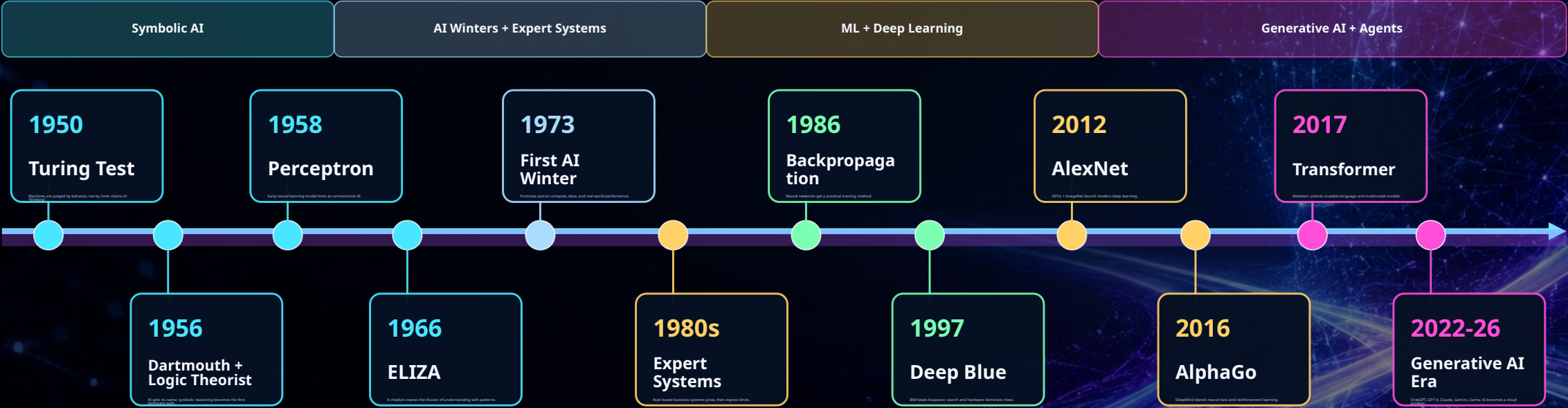
Indiana Wesleyan University | Machine Learning Fundamentals

1950

2026

AI/ML did not move in a straight line. It moved in waves.

Logic started the field. Winters reset expectations. Data, GPUs, cloud, transformers, and generative interfaces turned AI into everyday infrastructure.



Primary course sources used: Puthuon (2024), IEEE Computer Society, Our World in Data; additional milestone checks from Dartmouth, IBM, DeepMind, OpenAI, Google Research, Nature, Stanford HAI.

Foundations: logic, symbols, and early machine reasoning

1950**Turing proposes the imitation game**

Turing reframed the question from "Can machines think?" to whether a machine could imitate intelligent behavior in conversation.

1956**Dartmouth creates the field**

The Dartmouth workshop gave AI its name and organized research around the idea that learning and intelligence could be described precisely enough for machines.

1956**Logic Theorist**

Newell, Simon, and Shaw built a program that proved mathematical theorems, showing that symbolic search could model parts of reasoning.

1958**Rosenblatt's perceptron**

The perceptron offered an early learning model based on weighted inputs. It was a small but important bridge toward neural networks.

1966**ELIZA and language illusion**

Weizenbaum's ELIZA showed how simple pattern matching could feel conversational, even without true understanding.

Core idea

Early AI treated intelligence as symbols, rules, search, and problem solving.

Limit

Programs worked in controlled worlds but struggled with messy real-world context.

Why it matters

The same question remains today: does the model understand, or is it imitating?

AI winters: when promises outran evidence

1973**Lighthill Report**

A critical UK report argued that AI had not delivered the major impact promised. Funding and confidence dropped sharply.

Late 1970s**First AI Winter**

Symbolic systems lacked enough compute, data, and practical flexibility to meet expectations.

1980s**Expert systems boom**

Companies built rule-based systems that captured specialist knowledge in narrow domains. Useful, but brittle and expensive to maintain.

1987-93**Second AI Winter**

The expert-system and LISP-machine markets cooled. Business interest fell again when costs and limits became clearer.

Lesson from the winters

AI progress needs all three: better algorithms, more data, and more compute. Hype without deployment evidence eventually collapses.

1986**Backpropagation survives the cold**

Rumelhart, Hinton, and Williams showed how networks could learn internal representations by adjusting weights.

Cold periods did not stop research. They forced the field to become more practical.

Data + GPUs + cloud turn ML into modern AI

1997**IBM Deep Blue**

A chess supercomputer defeated Garry Kasparov, proving that brute-force search plus specialized hardware could beat elite human play in a narrow task.

2012**AlexNet breakthrough**

Krizhevsky, Sutskever, and Hinton used GPUs and deep convolutional networks to dominate ImageNet and start the deep-learning boom.

2009**ImageNet scale**

Large labeled image data made it possible to benchmark visual recognition and reward models that improved in the real world.

2016**AlphaGo wins 4-1**

DeepMind combined neural networks, search, and reinforcement learning to beat Lee Sedol in Go, a game long considered too complex for brute force.

3

ingredients converge

2012

deep learning inflection

4-1

AlphaGo vs. Lee Sedol

This era shifted AI from hand-written rules toward systems that learn patterns from data at scale.

Transformers and generative AI move AI into daily work

2017**Transformer architecture**

Google researchers introduced attention-based models that train in parallel and scale across language, images, and documents.

2020**GPT-3 scale signal**

Large language models showed that scaling parameters, data, and compute could produce broad language abilities.

2022**ChatGPT release**

OpenAI turned LLM interaction into a simple chat interface, making generative AI visible to everyday users.

2023**GPT-4 + multimodal shift**

Models began handling more complex reasoning tasks and multimodal inputs, while verification and safety became central concerns.

The interface changed the adoption curve.

Before ChatGPT, AI was often hidden behind apps. After ChatGPT, the user could directly ask the model for help.

2024-26**Agents, copilots, and model competition**

OpenAI, Google DeepMind, Anthropic, Meta, NVIDIA, and cloud providers push AI into coding, search, productivity, analytics, and business workflows.

The core challenge changed too.

Teams now need accuracy checks, privacy controls, bias testing, cost monitoring, and human judgment.

What the timeline means for work now

AI moved from lab demonstrations to cloud-backed products embedded in everyday workflows.

Vision

medical imaging, inspection, autonomous systems

Language

search, translation, support, documentation

Coding

copilots, testing, debugging, app modernization

Business

risk scoring, forecasting, CRM, analytics

Robotics

factory automation, warehouse systems, embodied AI

Governance

privacy, bias, transparency, security, accountability

Leadership point:

Good AI adoption is not just model choice. It is data quality, workflow fit, human review, and clear accountability.

Sources used to build the timeline

Course sources

- Putchuon - "Put You On." (2024). The entire history of artificial intelligence (Last 100 years) [Video]. YouTube.
- IEEE Computer Society. (n.d.). The evolution of AI: From foundations to future prospects.
- Roser, M. (2022). The brief history of artificial intelligence. Our World in Data.

Additional checks

Dartmouth AI history; Turing (1950); CMU Simon Initiative; Cornell perceptron article; Nature backpropagation paper; IBM Deep Blue history; Google Research Transformer paper; DeepMind AlphaGo page; OpenAI ChatGPT/GPT-4 pages; Stanford HAI AI Index/FMTI.

AI tools used

I used ChatGPT to help research, organize milestones, draft concise descriptions, and design the visual structure. I used DALL·E to create the cinematic background images. I reviewed and edited the final timeline content for accuracy and clarity.

Originality note

This timeline layout, wording, and visual design were created as an original submission for AIML-500 Workshop 1.3.

Discussion post line to paste:

Attached is my original AI/ML timeline. I used ChatGPT for research support and DALL·E for visual background assets, then reviewed and edited the final timeline myself.